

Predictive Model Plan

1. Model Logic (Generated with GenAI)

Objective

The objective of this predictive model is to identify customers who are at risk of becoming delinquent and missing future payments. The model will use historical customer financial and behavioral information to generate a delinquency risk prediction.

Selected Model

Logistic Regression

Dataset Summary

- Total Records: 500
- Total Variables: 19
- Delinquent Customers: 80
- Non-Delinquent Customers: 420

Missing Values Identified

- Income: 39 missing values
- Loan_Balance: 29 missing values
- Credit_Score: 2 missing values

Top 5 Input Features

1. Credit_Utilization
2. Missed_Payments
3. Debt_to_Income_Ratio
4. Credit_Score
5. Loan_Balance

Model Workflow

1. Load customer dataset.
2. Handle missing values using median imputation.
3. Select key risk-related features.
4. Split data into training and testing datasets.
5. Train a Logistic Regression classification model.
6. Generate probability scores for each customer.
7. Classify customers as high-risk or low-risk.
8. Provide risk predictions to support proactive intervention and collections strategies.

The model is designed to transform customer financial characteristics into a probability score indicating the likelihood of future delinquency.

2. Justification for Model Choice

Logistic Regression was selected because the target variable, Delinquent_Account, is binary (0 = Non-Delinquent, 1 = Delinquent). The dataset contains structured financial and behavioral variables that are well suited for Logistic Regression. The model provides probability-based predictions, enabling Geldium to rank customers by risk level and prioritize intervention efforts.

A major advantage of Logistic Regression is its interpretability. Financial institutions must understand and justify model decisions due to regulatory requirements and customer fairness considerations. Unlike neural networks, Logistic Regression clearly shows how variables such as credit utilization, missed payments, debt-to-income ratio, and credit score influence delinquency risk. The model is also computationally efficient, easy to deploy, and widely used in credit risk assessment.

3. Evaluation Strategy

The model will be evaluated using several performance metrics to ensure accuracy, reliability, and fairness.

Accuracy

Measures the overall percentage of correct predictions.

Precision

Measures how many customers predicted as delinquent are actually delinquent.

Recall

Measures how many actual delinquent customers are correctly identified. Recall is particularly important because missing high-risk customers could result in financial losses.

F1 Score

Provides a balanced assessment of precision and recall.

AUC-ROC

Measures the model's ability to distinguish between delinquent and non-delinquent customers across different probability thresholds.

Confusion Matrix

Provides a detailed breakdown of prediction outcomes and helps identify false positives and false negatives.

Bias and Fairness Assessment

The model will be evaluated for potential bias arising from historical patterns, class imbalance, or proxy variables. Since the dataset contains 420 non-delinquent customers and 80 delinquent customers, class imbalance should be monitored to ensure fair predictions.

Fairness checks will include:

- Monitoring performance across customer groups.
- Reviewing feature selection to avoid proxy bias.
- Conducting periodic model audits.
- Maintaining human oversight over AI-assisted decisions.

Ethical Considerations

The model should support decision-making rather than replace human judgment. Customer financial information must be handled responsibly, predictions should remain explainable, and fairness should be continuously monitored to prevent unintended discrimination.

Conclusion

Based on the dataset characteristics and business requirements, Logistic Regression is the most appropriate model for predicting customer delinquency. The model offers transparency, explainability, and reliable performance while supporting proactive risk management. By combining predictive analytics with fairness monitoring and human oversight, Geldium can improve customer risk assessment and reduce delinquency-related losses.